

Habtamu Asefa

AI Research Engineer

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AI research engineer working on Amharic speech models, with production experience in Python, TypeScript, React Native, and backend services. Founder of PrepX.

EXPERIENCE

Liqawnt | AI Research & Engineering 2026-present

- Develop an Amharic TTS pipeline for text normalization, audio preparation, synthesis, and evaluation.
- Implemented VITS encoder-side fine-tuning in PyTorch with monotonic alignment search, freezing the waveform decoder to reduce training memory.

PrepX | Founder & Lead Engineer 2024-present

- Built and released a study app for Ethiopian Grade 9-12 students; reached **2,600+ downloads and 93,173+ questions answered** following the March 2026 public launch.
- Developed React Native and Expo study flows, offline downloads, streamed AI chat, and live competitions; built FastAPI and NestJS services for content, subscriptions, and notifications.
- Integrated textbook and past-exam retrieval with AI tutors using Qdrant and FastEmbed; use Typesense for content search across **1,824 curriculum topics**.

Freelance & Agency | Software Engineer 2023-present

- Delivered React Native features for Dymnd, including timezone-sensitive habit tracking and fixes to authentication, consent, keyboard behavior, and navigation.
- Maintained backend services and deployment infrastructure for a US bar-exam preparation platform using Node.js, AWS EC2, Nginx, and TLS certificate automation.

Temaribet | Founder 2019-2025

- Founded and operated a tutoring marketplace; managed tutor onboarding and family matching. Selected into Reach for Change and Orbit Innovation Hub.

SELECTED TECHNICAL WORK

react-native-ajora | Author & Maintainer 2025-present

Published a React Native library with AG-UI integration, streamed messages, tool-call rendering, and client-side tools. The package is used in PrepX.

myTorch | Learning Project 2026

Implemented scalar reverse-mode autograd, MLP layers, SGD, and a byte-level BPE tokenizer in Python; added tests for graph and gradient behavior.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

ML: PyTorch, torchaudio, Hugging Face Transformers

Applications: React Native, Expo, FastAPI, NestJS

Data: PostgreSQL, SQLite, Qdrant, Typesense

Infrastructure: Linux, Docker, Nginx, AWS EC2, S3, GitHub Actions

EDUCATION

Addis Ababa University / AAiT 2024

BSc in Biomedical Engineering. Thesis: UlcerGuard, pressure-sensing footwear for diabetic foot-ulcer prevention.

PrepX metrics: October 5, 2026.